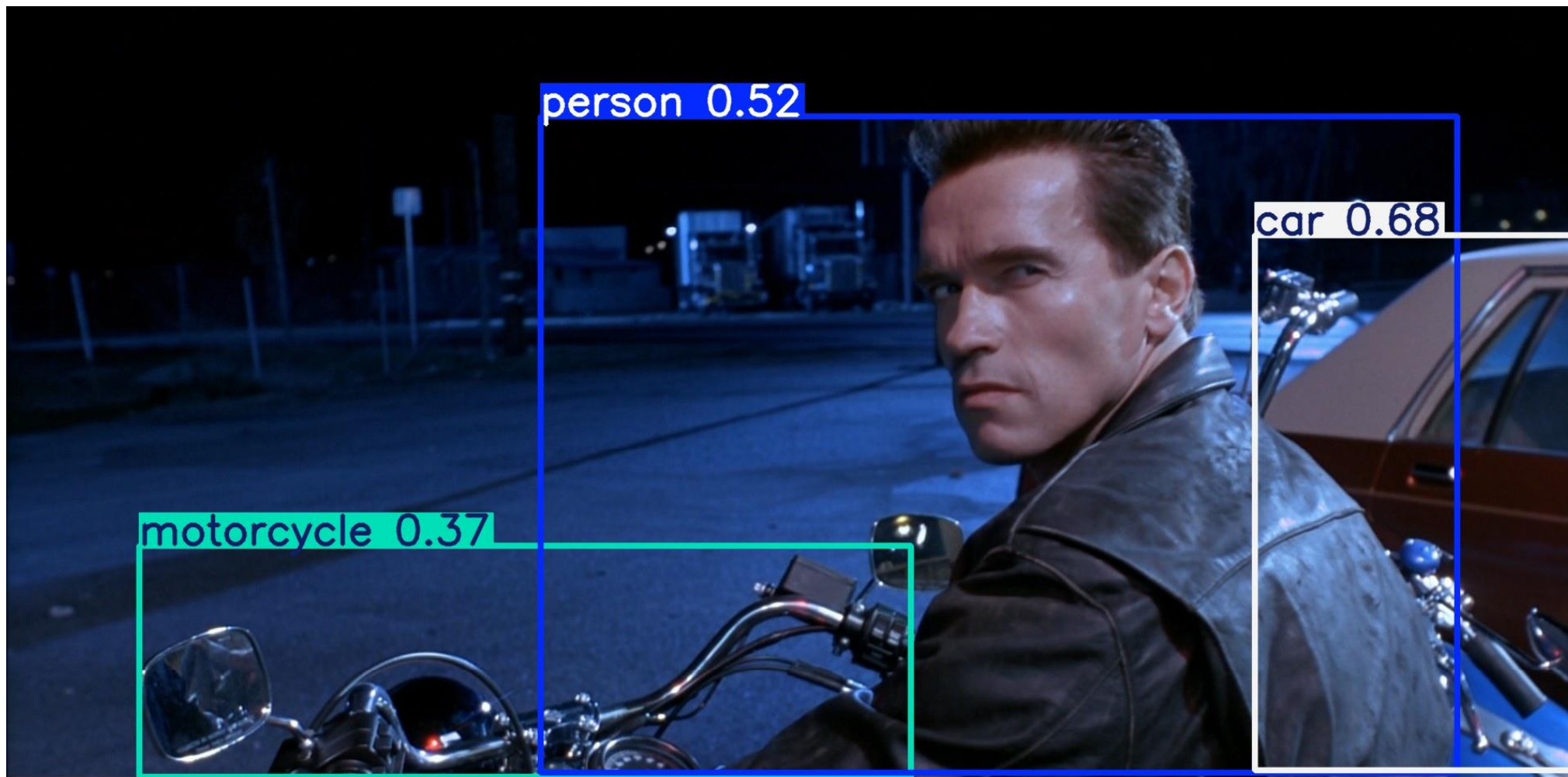


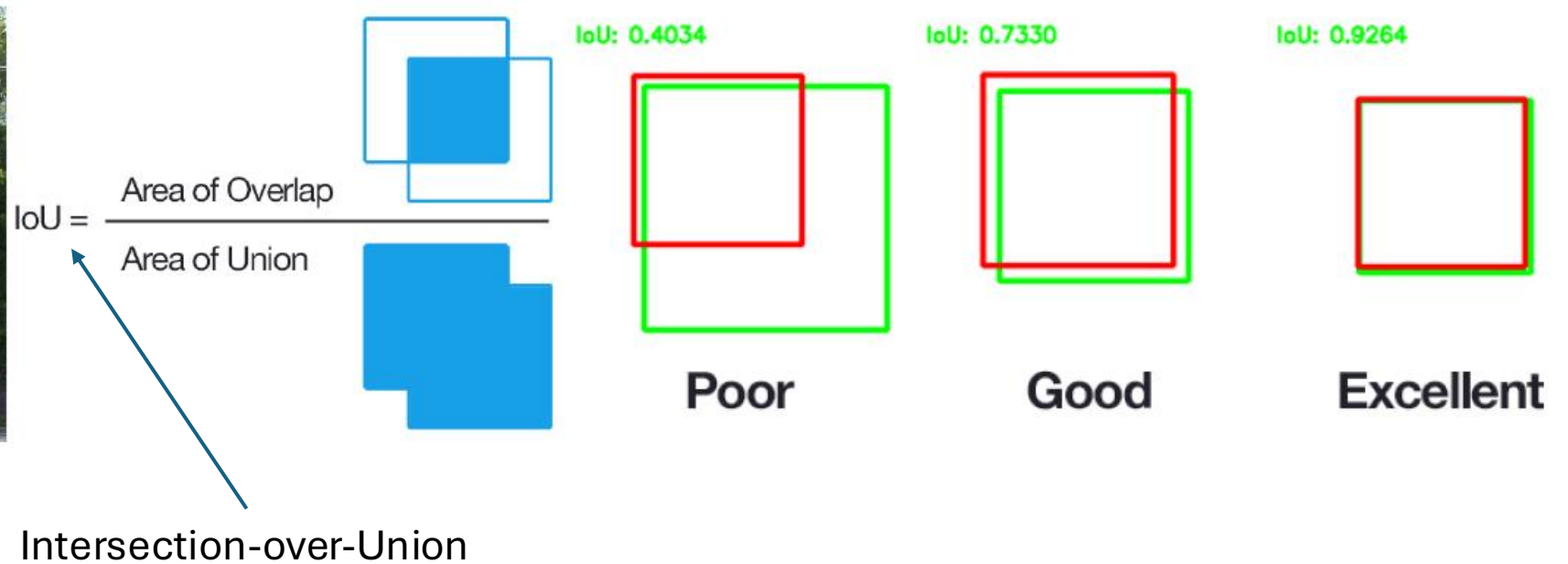
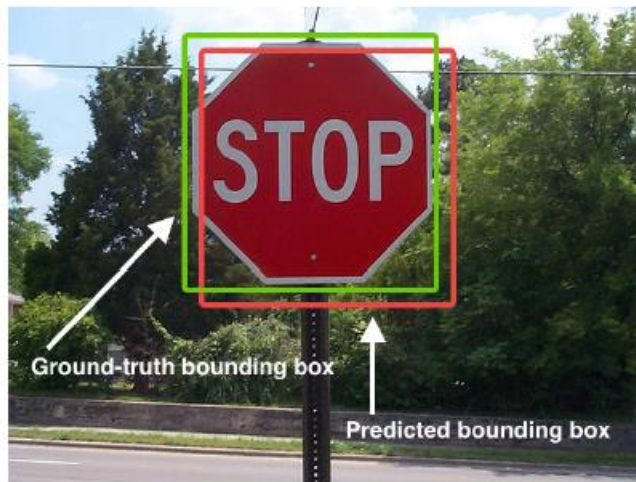
Object Detection

Deep Learning and Image Processing

output: bounding boxes with category label and confidence



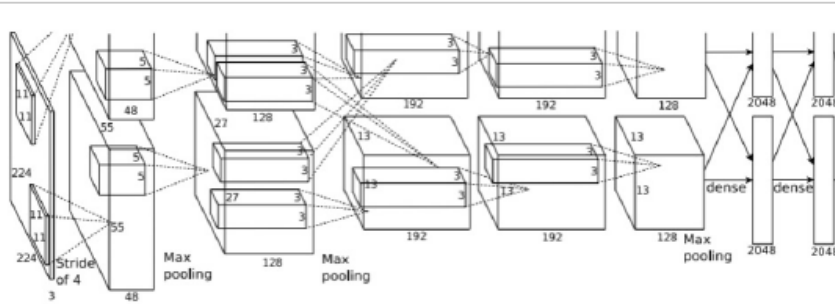
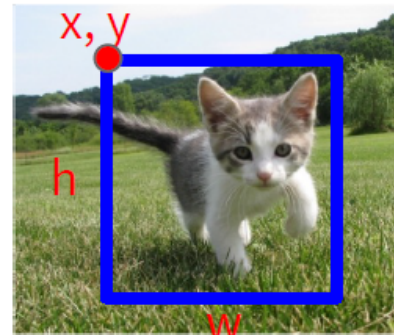
Performance Evaluation of Localization



images with multiple objects: number of detections dependent on used confidence threshold

Object Detection: Single Object

(Classification + Localization)



Vector:
4096

Fully
Connected:
4096 to 1000

Class Scores

Cat: 0.9
Dog: 0.05
Car: 0.01
...

Correct label:
Cat

Softmax
Loss

Multitask Loss

Fully
Connected:
4096 to 4

Box
Coordinates
(x, y, w, h)

L2 Loss

Correct box:
(x', y', w', h')

Treat localization as a
regression problem!

too complicated for multiple objects

Localization for Multiple-Object Images

idea: classify many different crops of the image as object or background
(crops: sliding window over image, iterated at multiple window sizes)



Dog? No
Cat? No
Background? Yes



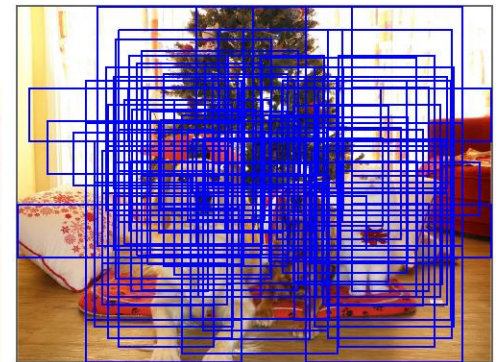
Dog? Yes
Cat? No
Background? No



Dog? Yes
Cat? No
Background? No



Dog? No
Cat? Yes
Background? No



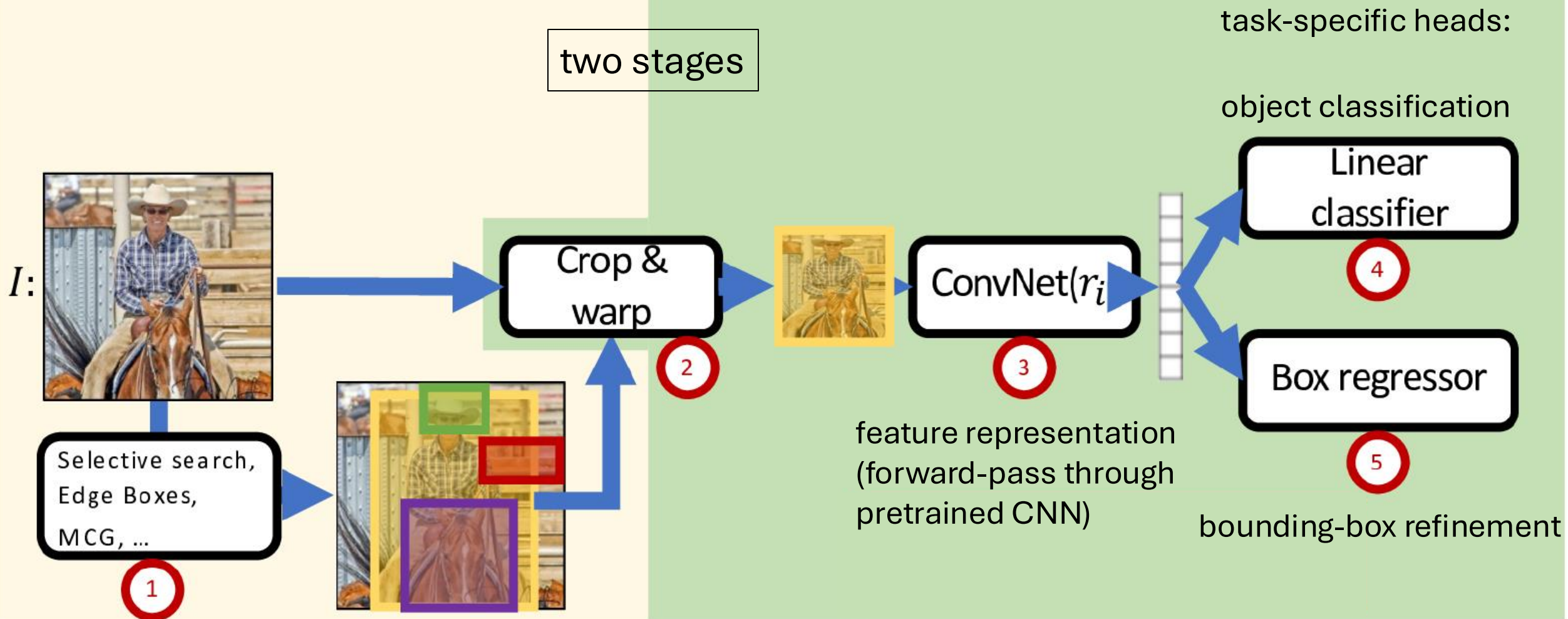
issue: too many
possible crops

→ need for region proposals

Region-Based Convolutional Neural Network (R-CNN)

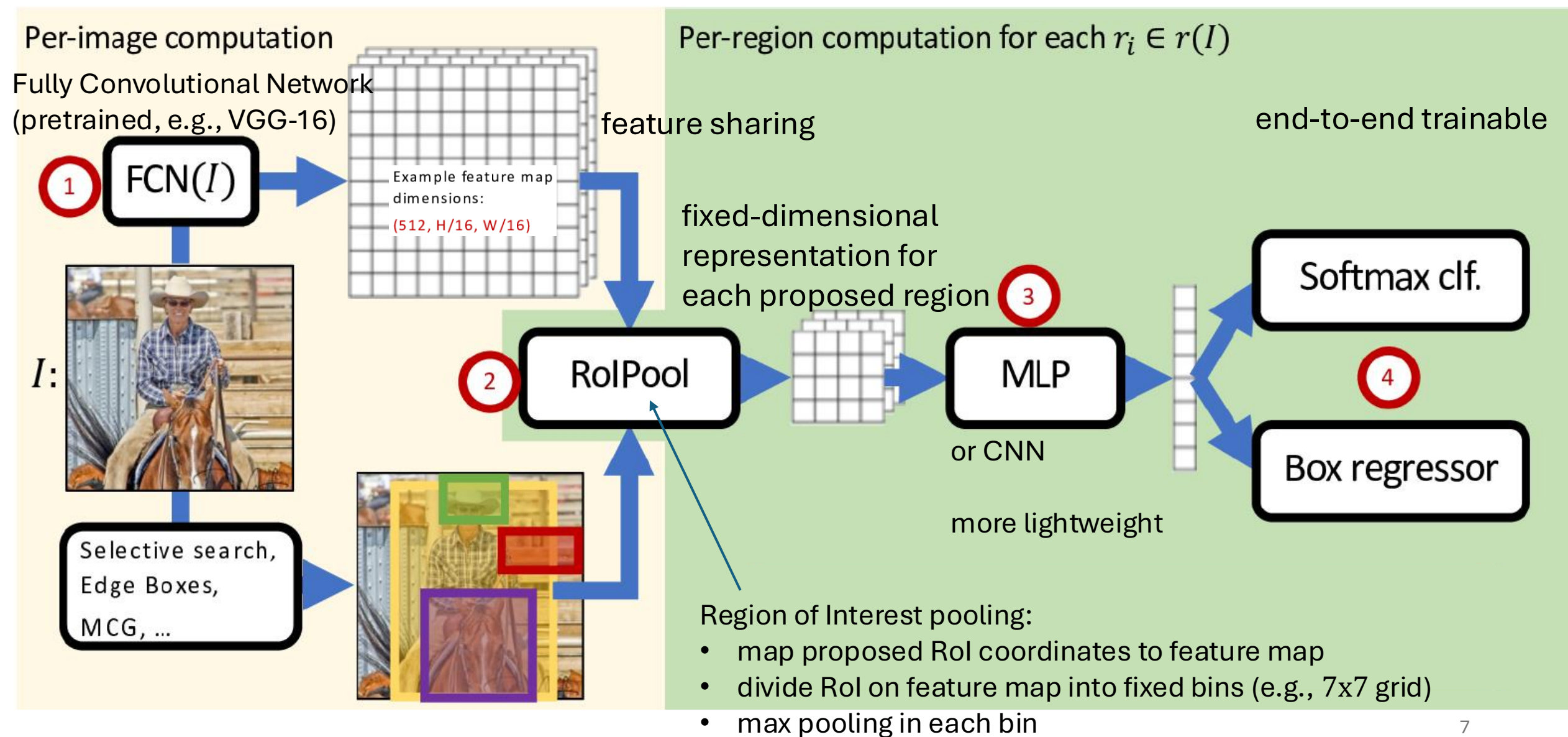
Per-image computation

Per-region computation for each $r_i \in r(I)$



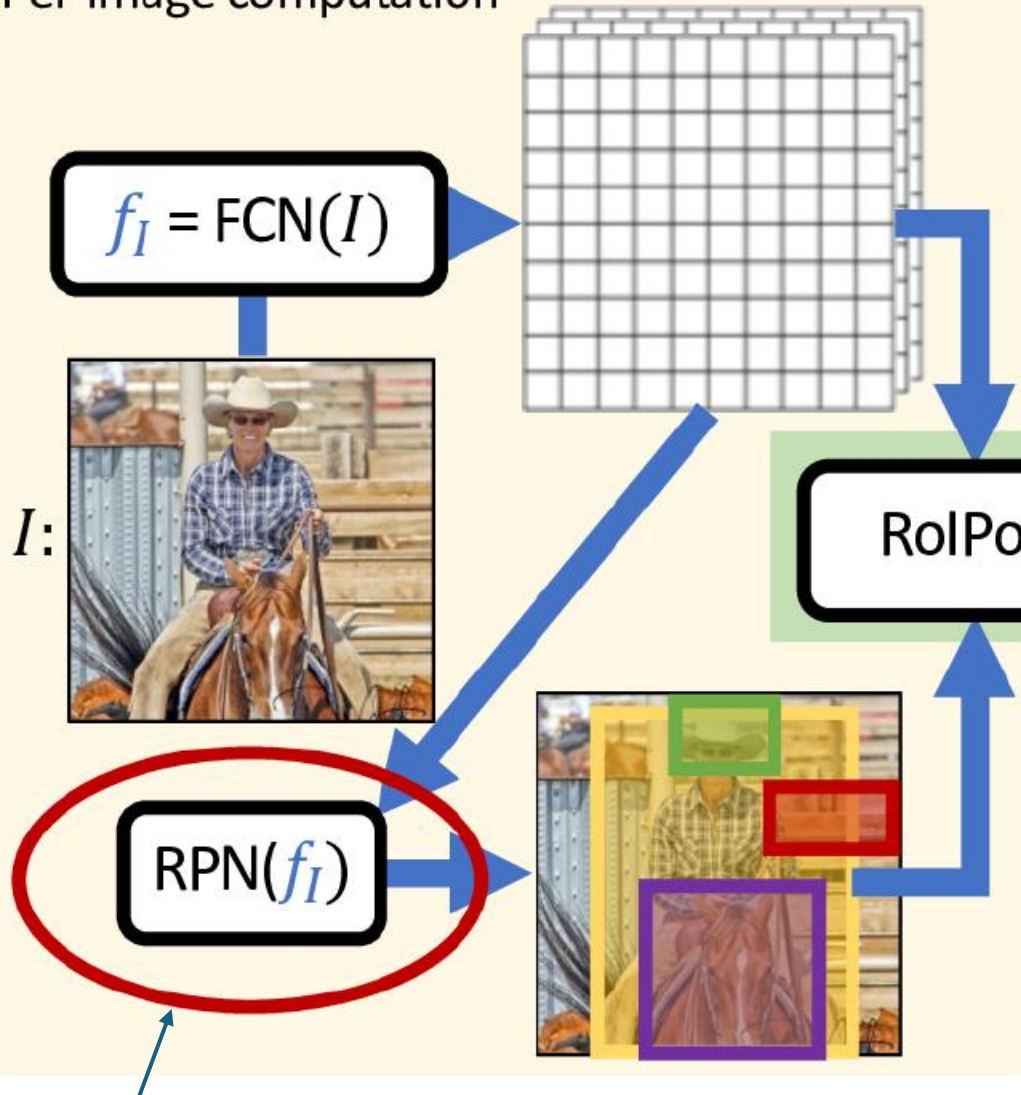
region proposals by "classic" method ($\sim 2k$)

Fast R-CNN: Crop ConvNet Features



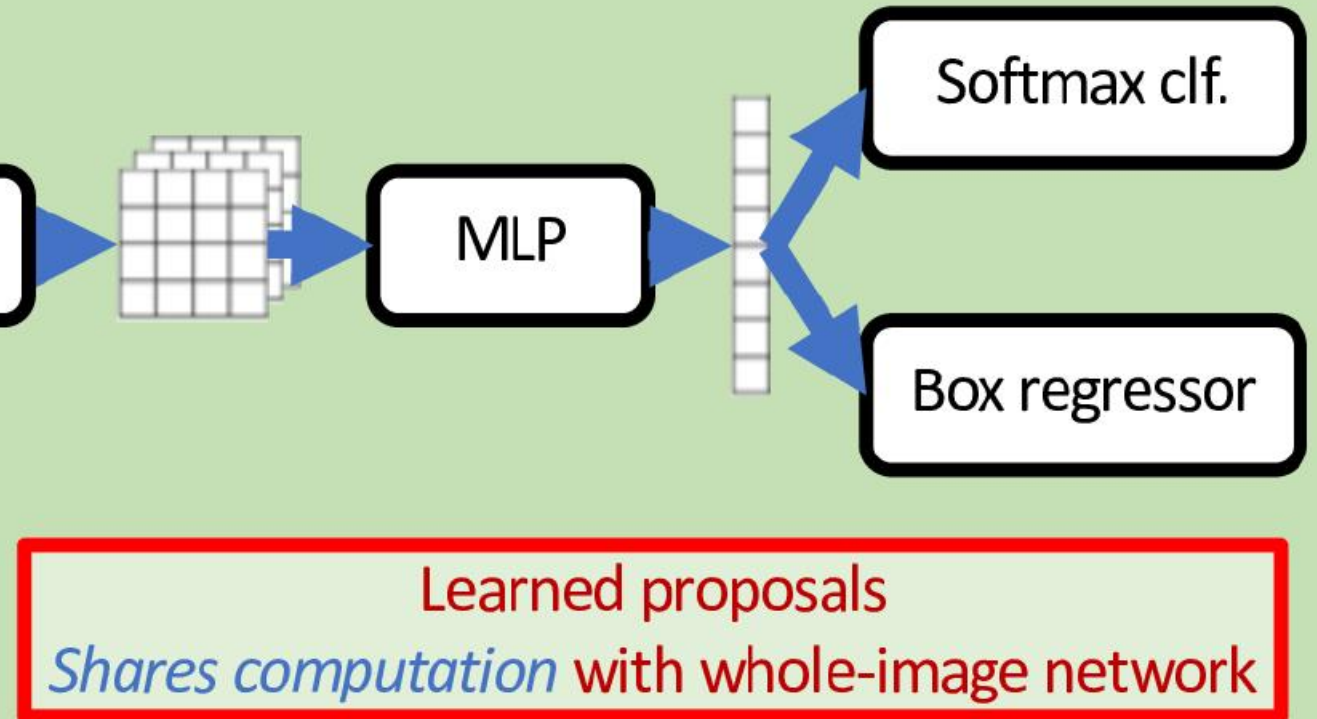
Faster R-CNN: Use Region Proposal Network

Per-image computation



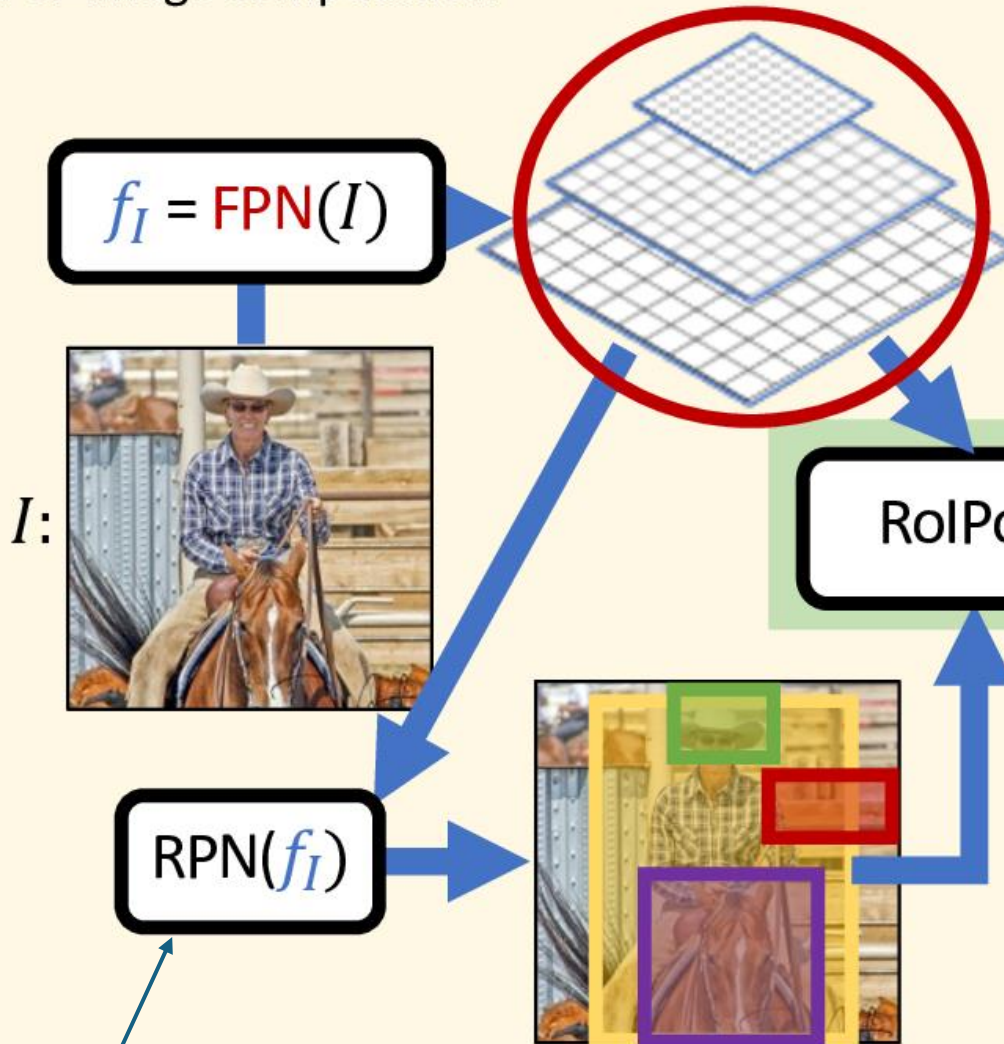
Per-region computation for each $r_i \in r(I)$

two RPN losses combined with two detection head losses during end-to-end training

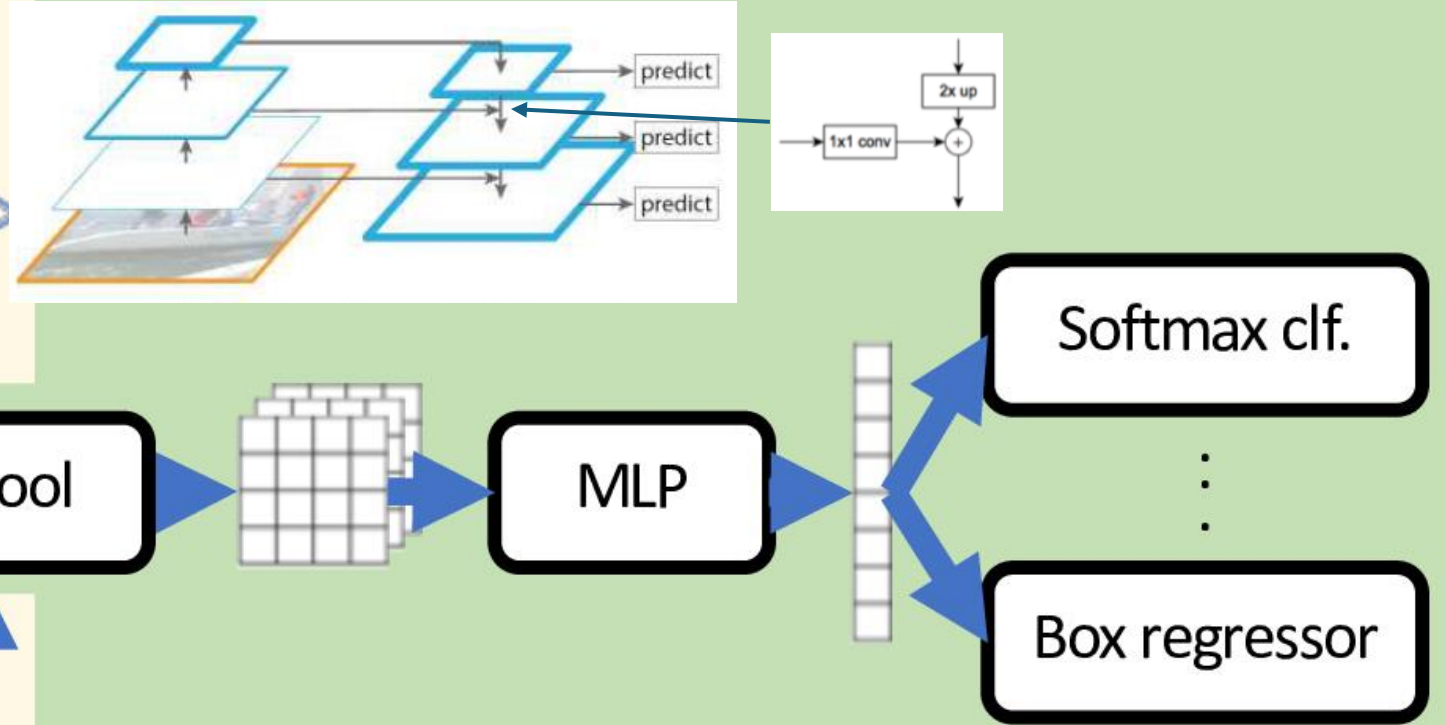


Feature Pyramid Network (FPN)

Per-image computation



Per-region computation for each $r_i \in r(I)$



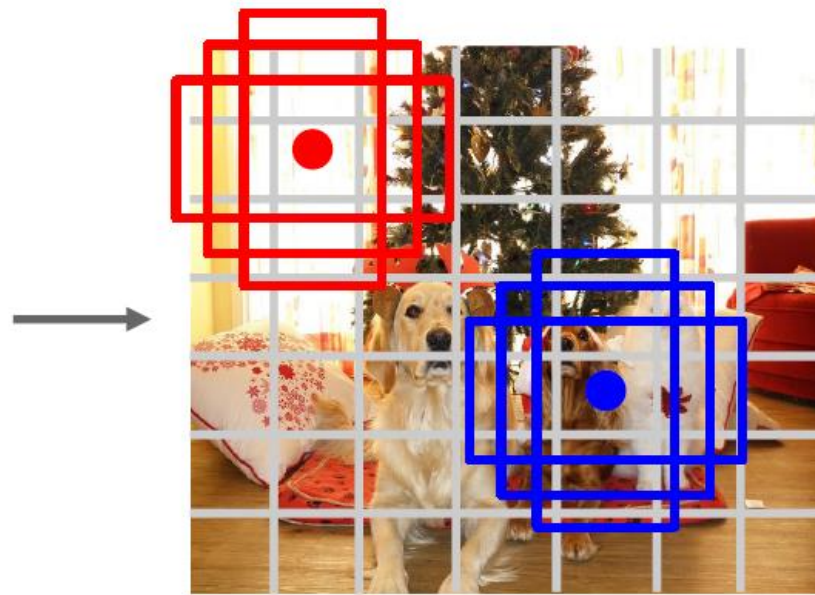
The whole-image feature representation can be improved by making it *multi-scale*

using different anchor sizes for different pyramid levels

Single-Stage Detectors: Drop Per-Region Computation



Input image
 $3 \times H \times W$



Divide image into grid
 7×7

Image a set of base boxes
centered at each grid cell
Here $B = 3$

Within each grid cell:

- Regress from each of the B base boxes to a final box with 5 numbers:
($dx, dy, dh, dw, confidence$)
- Predict scores for each of C classes (including background as a class)
- Looks a lot like RPN, but category-specific!

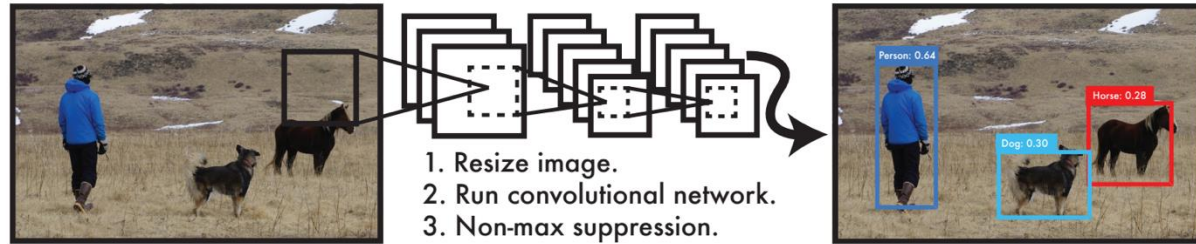
Output:
 $7 \times 7 \times (5 * B + C)$

Redmon et al, "You Only Look Once:
Unified, Real-Time Object Detection", CVPR 2016
Liu et al, "SSD: Single-Shot MultiBox Detector", ECCV 2016
Lin et al, "Focal Loss for Dense Object Detection", ICCV 2017

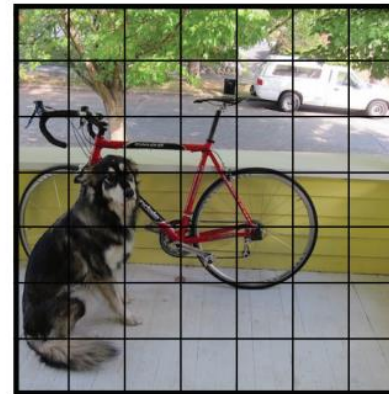
faster, but usually worse performance

YOLO: Real-Time Object Detection

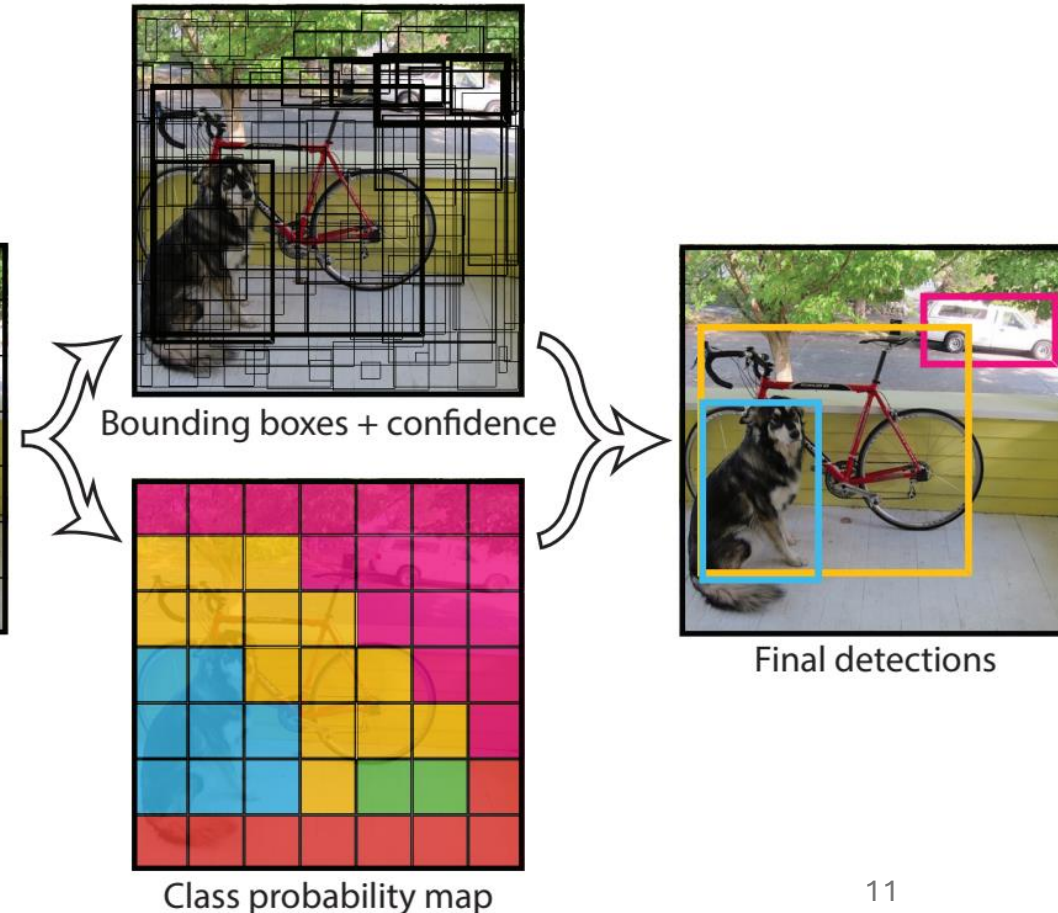
anchor-free approach
in newer YOLO versions



You Only Look Once:
prediction of bounding boxes
and class probabilities in one go



$S \times S$ grid on input



Object Tracking

task: locating an object in successive video frames

basic approach:

1. detection: localize target object(s) (e.g., YOLO)
2. motion estimation: predict next location (e.g., Kalman filter or optical flow)
3. appearance matching: comparison of previous and predicted next location (e.g., feature descriptor, CNN features)

detection repeated frequently to correct for accumulated deviations

Optical Flow

dense optical flow:

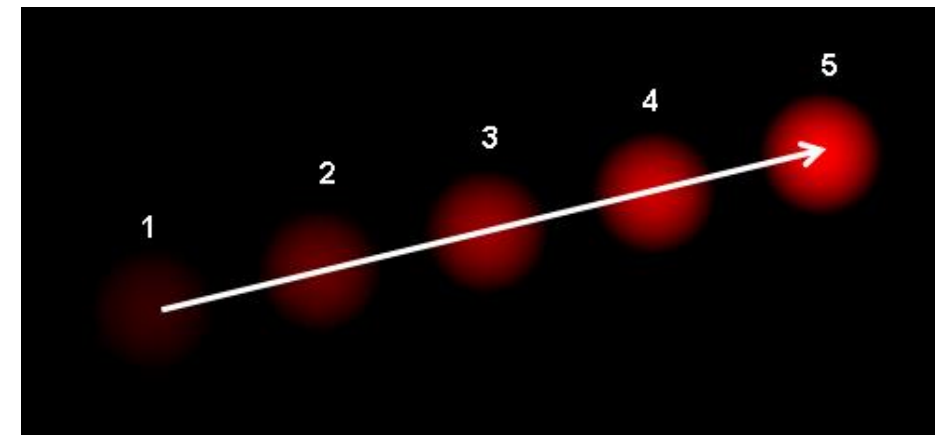
estimate motion vector of every pixel

e.g., Horn-Schunck method or Lucas-Kanade (LK) method

sparse optical flow:

estimate motion vector of few image features

e.g., Kanade-Lucas-Tomasi (KLT) feature tracker



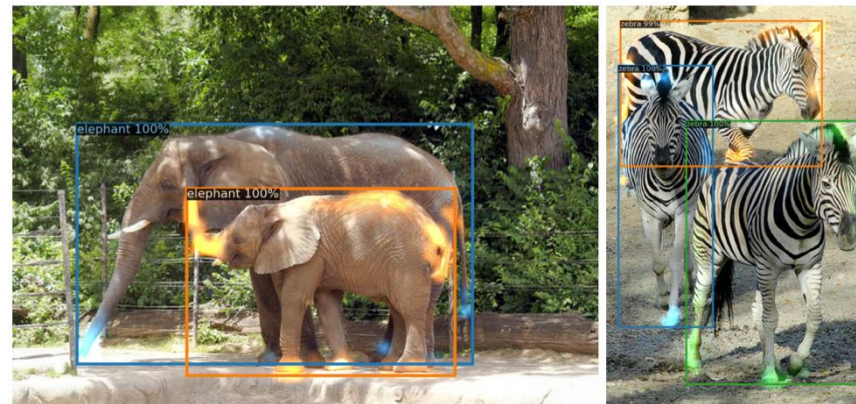
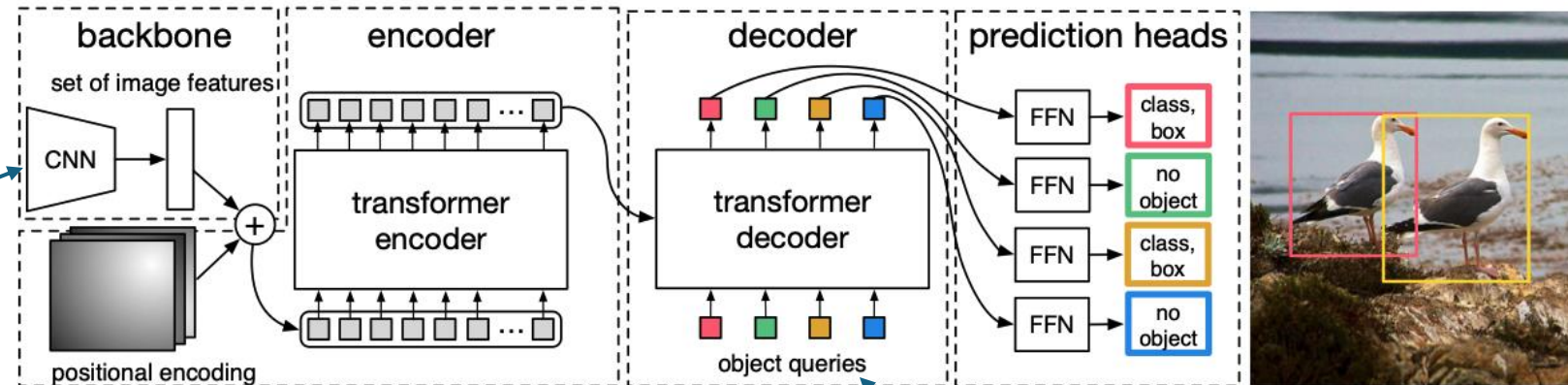
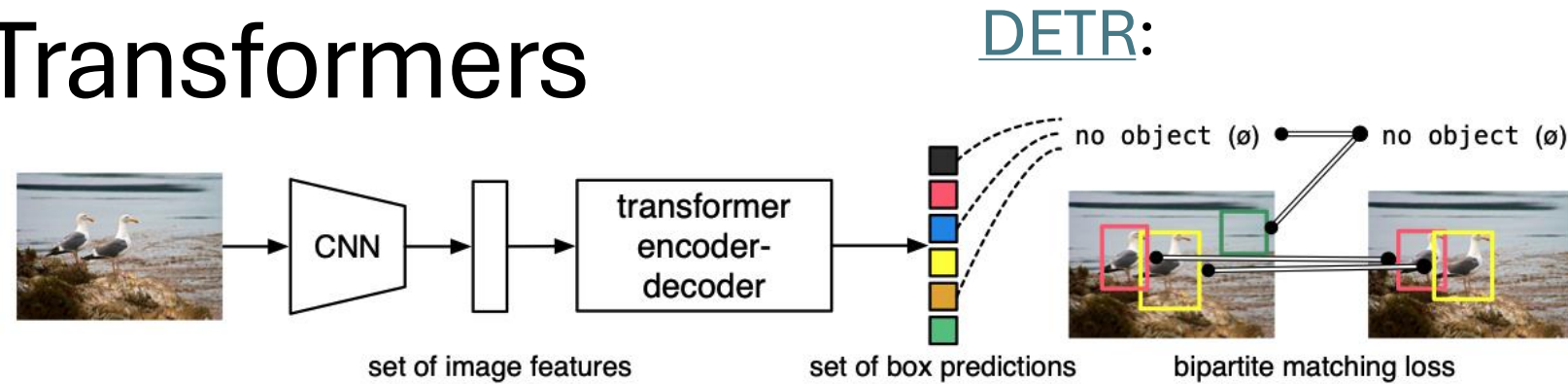
Detection with Transformers

idea: use a transformer to directly predict object bounding boxes and class labels from image features (no RPN)

backbone can also be vision transformer

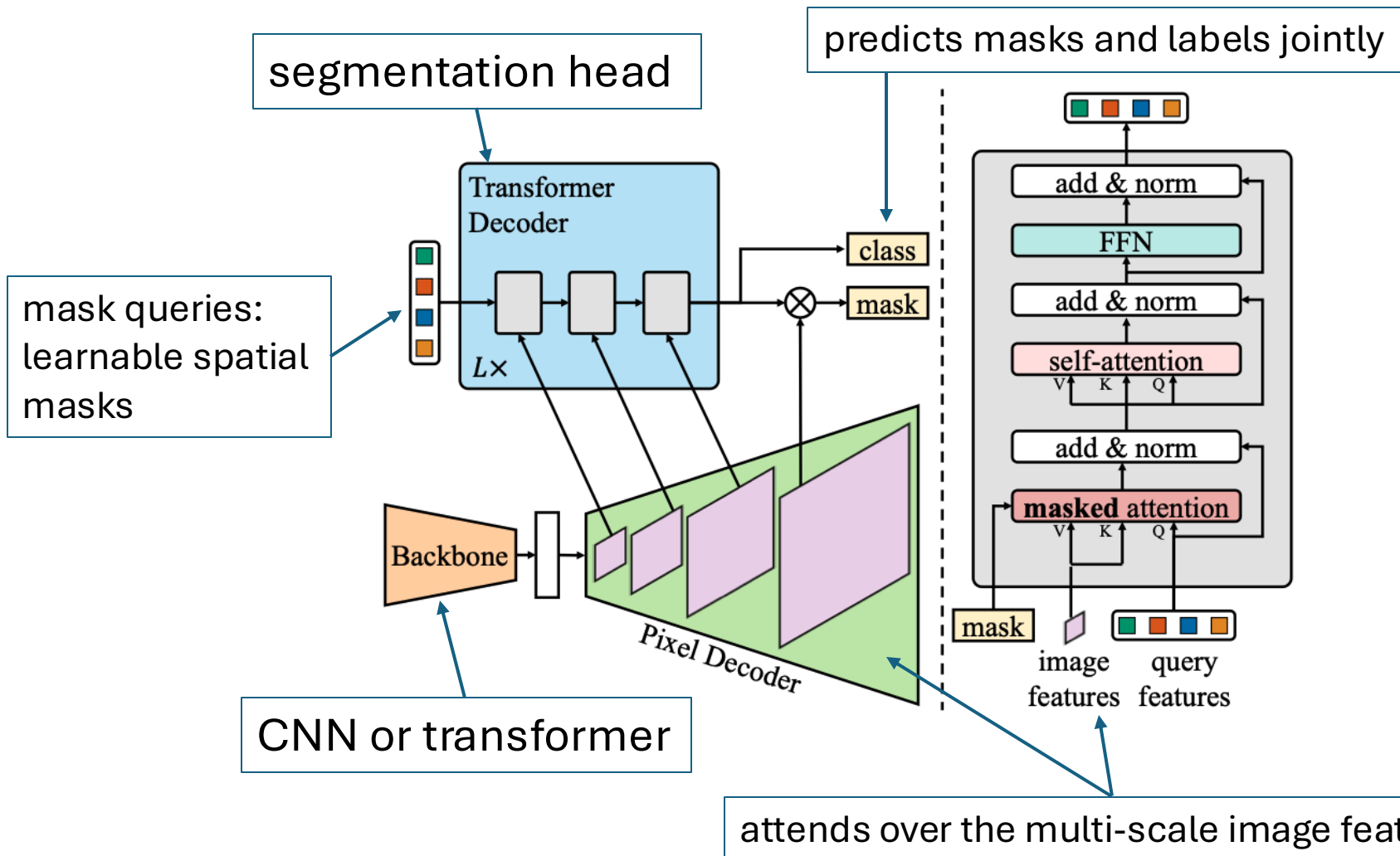
→ popular combination: DINOv2 + DETR

visualized decoder attention for predicted objects:



slot for learnable vectors
("Is there an object here?")

Segmentation with Transformers



Mask2Former

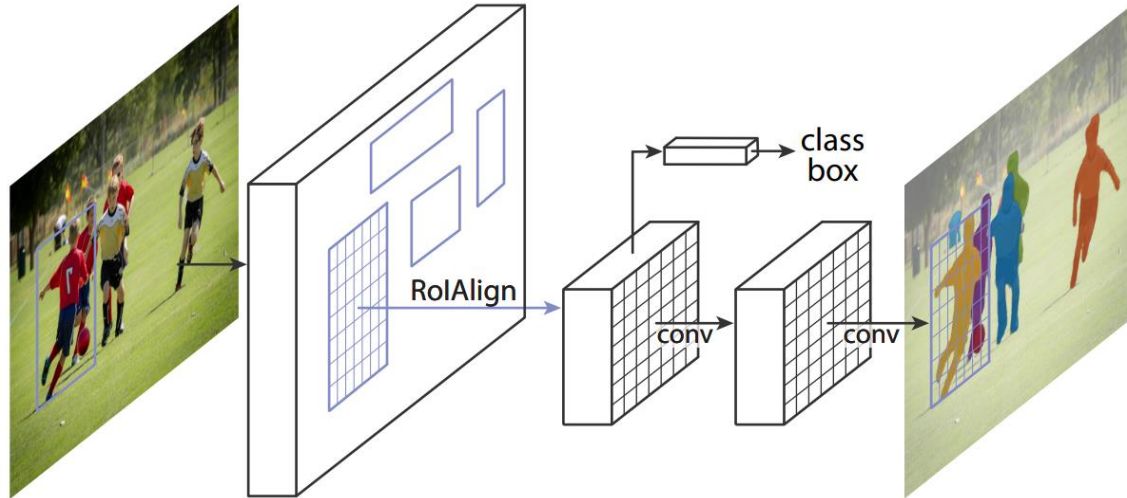
unlike pixel-wise classification, it predicts a set of masks, each tied to an object or region

(similar to DETR, but predicting segmentation masks instead of bounding boxes)

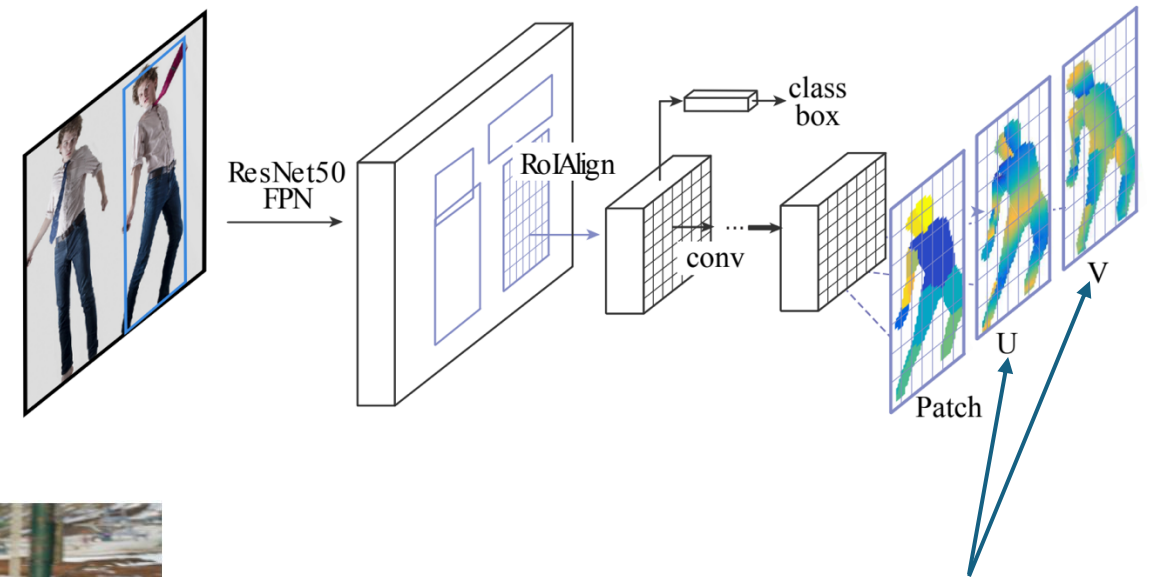
Instance Segmentation

Add Additional Network Heads to R-CNN

instance segmentation ([Mask R-CNN](#)):



pose estimation ([DensePose](#)):

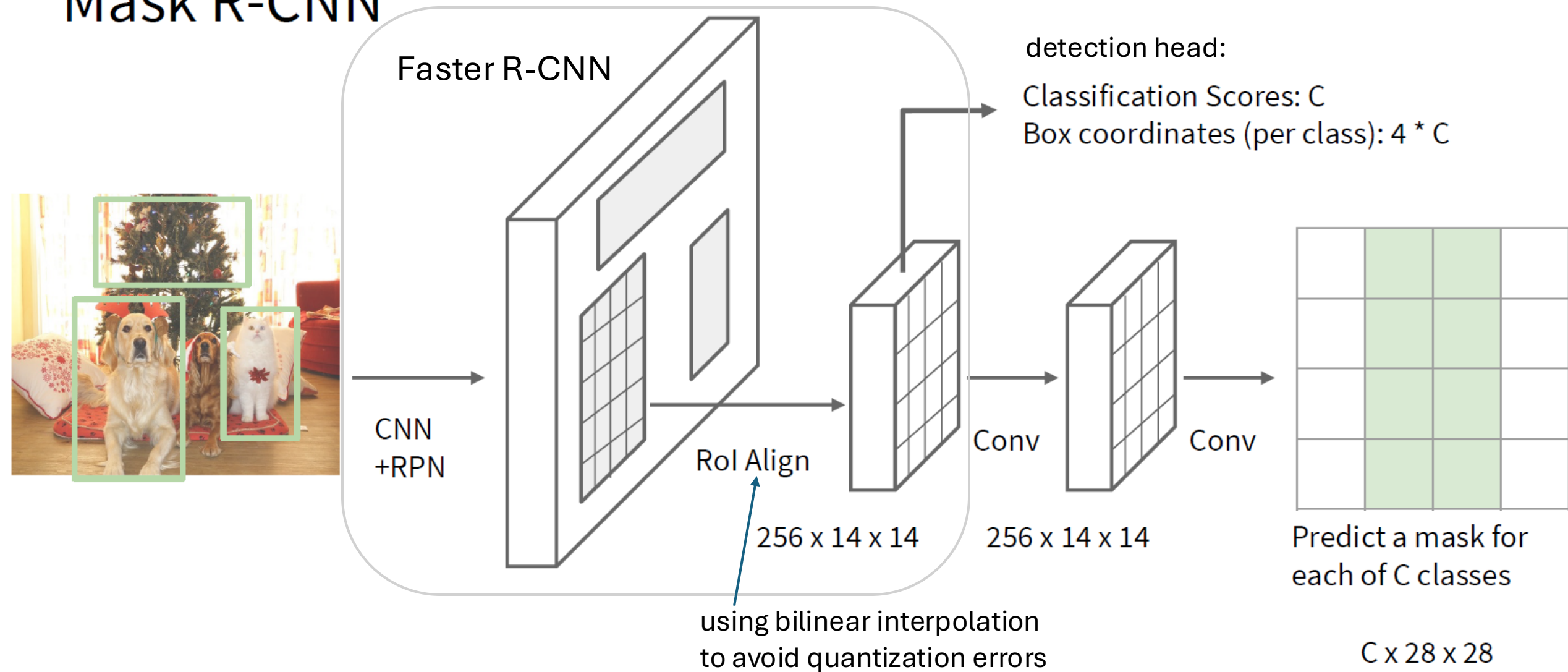


3D model's surface to 2D



Mask R-CNN

→ total loss is sum of RPN, detection-head, and mask-head losses



1. object detection (boundary boxes)
2. prediction of separate binary masks for each detected object (specific instances)¹⁸

image with training proposal

28 × 28 mask target

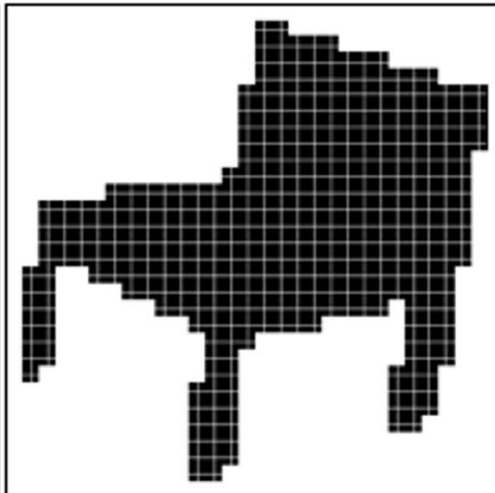
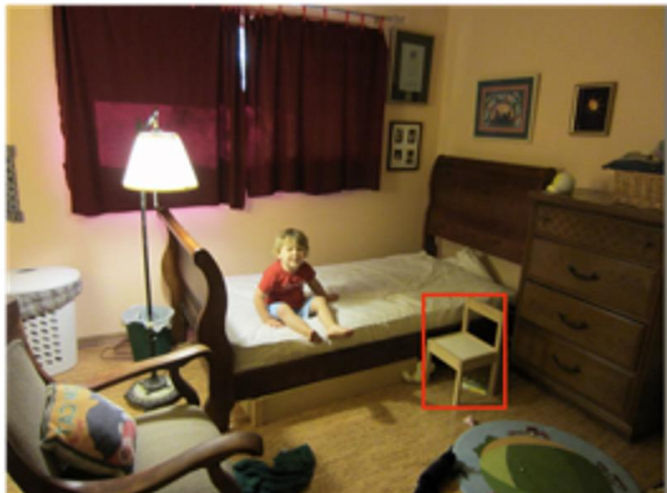
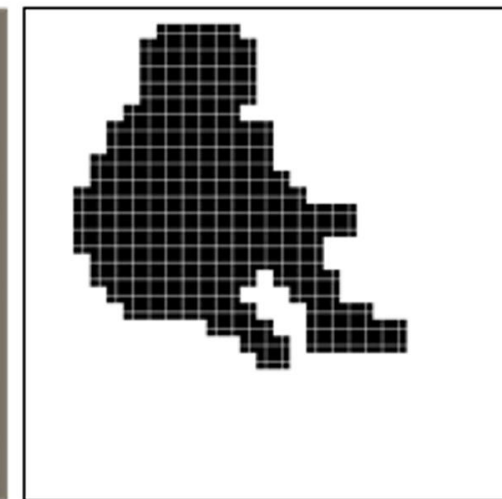
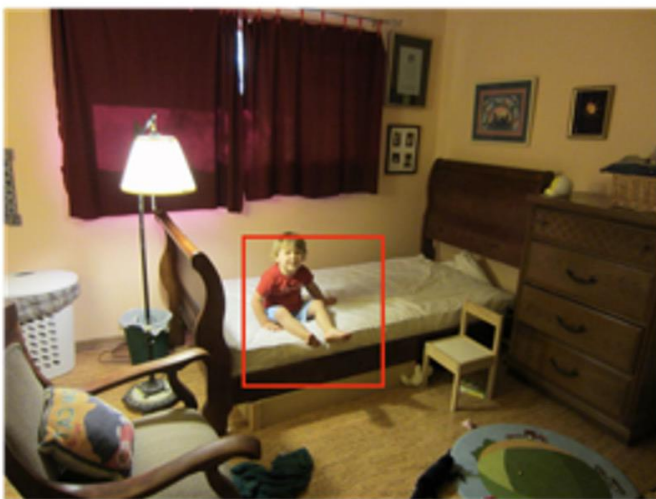
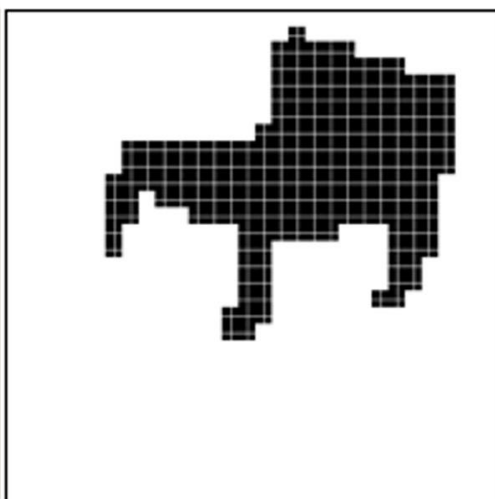
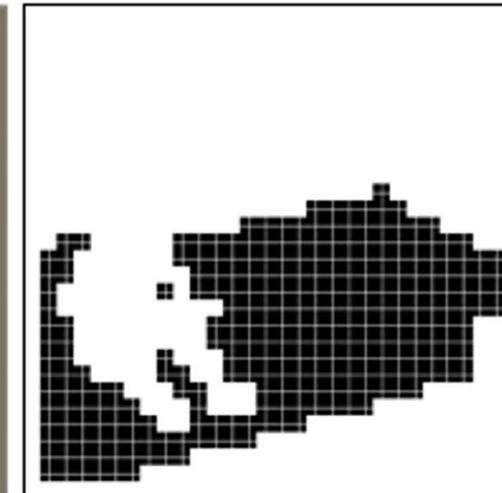
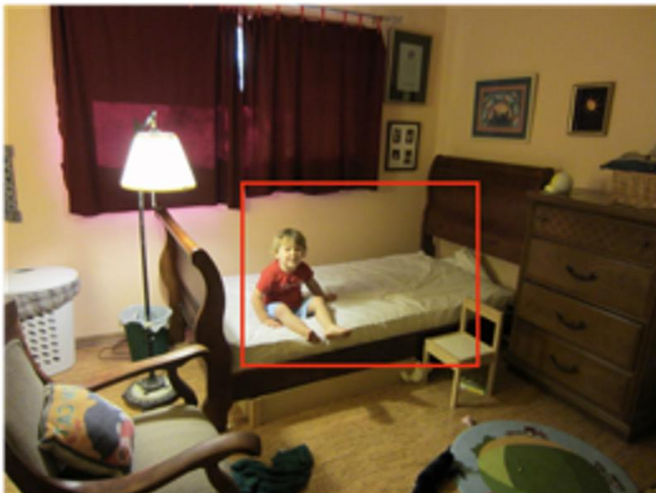
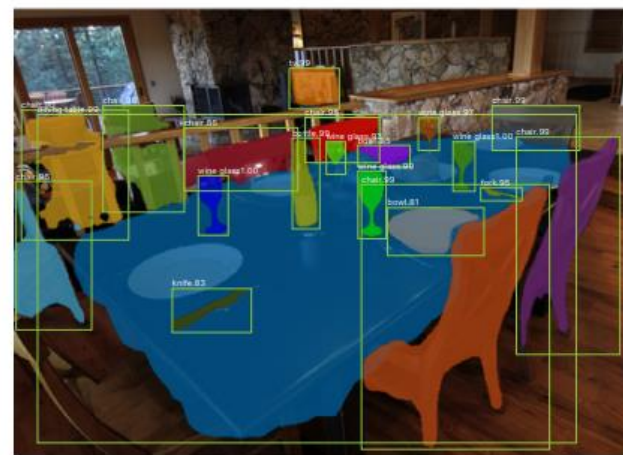
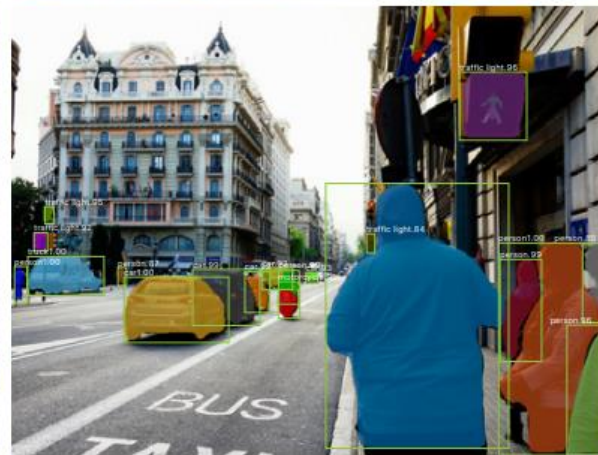
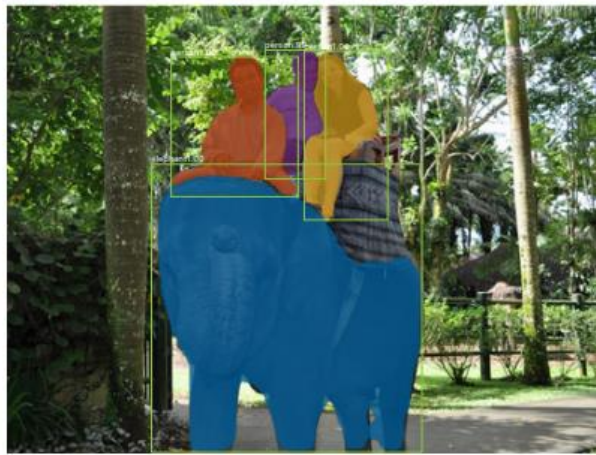
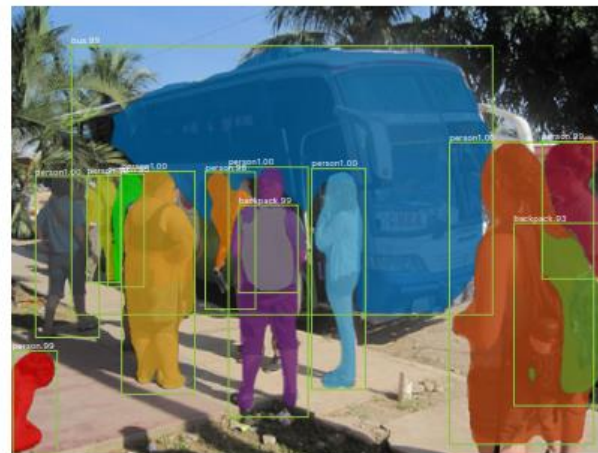
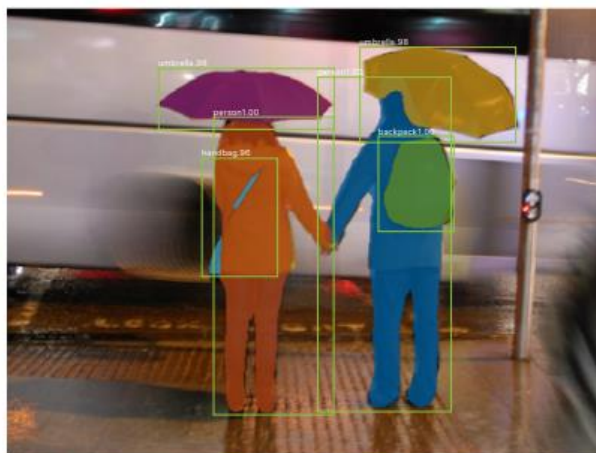
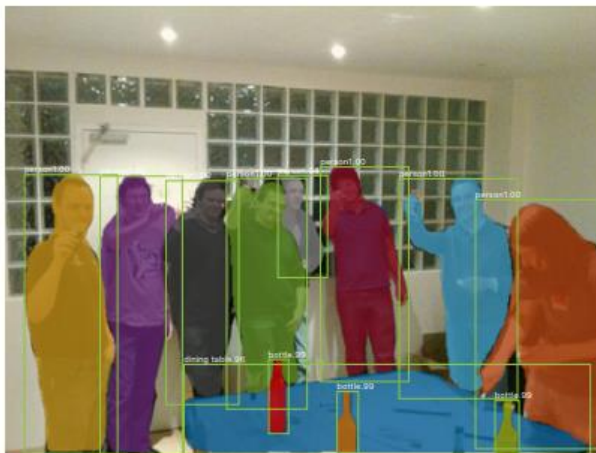


image with training proposal

28 × 28 mask target



results on MS COCO data set



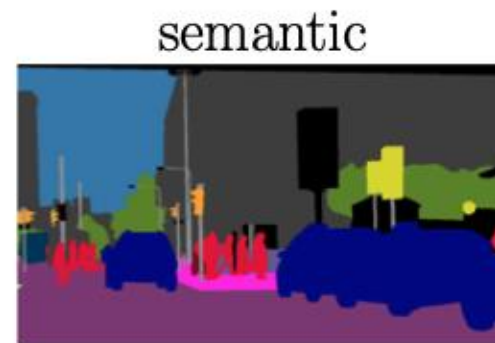
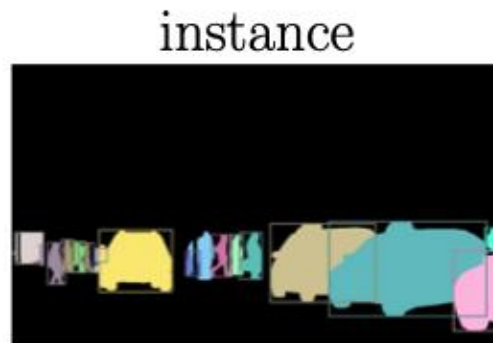
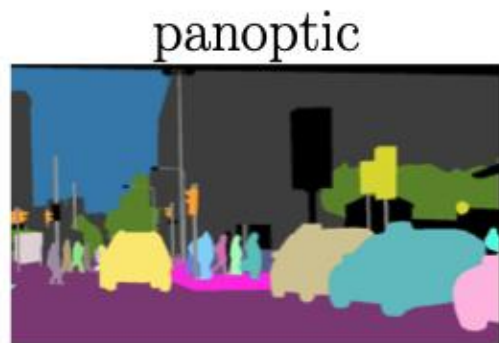
Panoptic Segmentation

unification of semantic segmentation (labeling all pixels) and instance segmentation (identifying individual objects)

each pixel is assigned

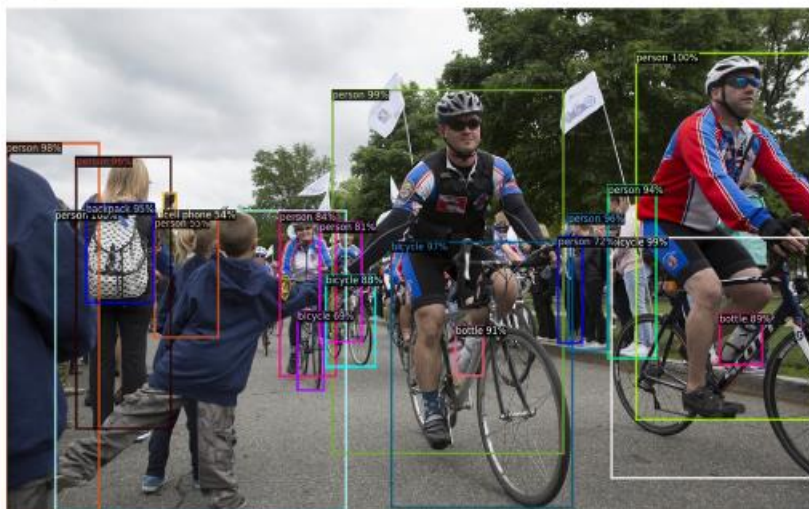
- a class label (e.g., road, car)
- an instance ID, if it belongs to a “thing” class (e.g., car #1, car #2)

covering both “thing” classes (countable objects) and “stuff” classes (uncountable regions like sky, grass, road)



e.g., with combination of Mask R-CNN and fully-convolutional head (or DETR variants such as Mask2Former)

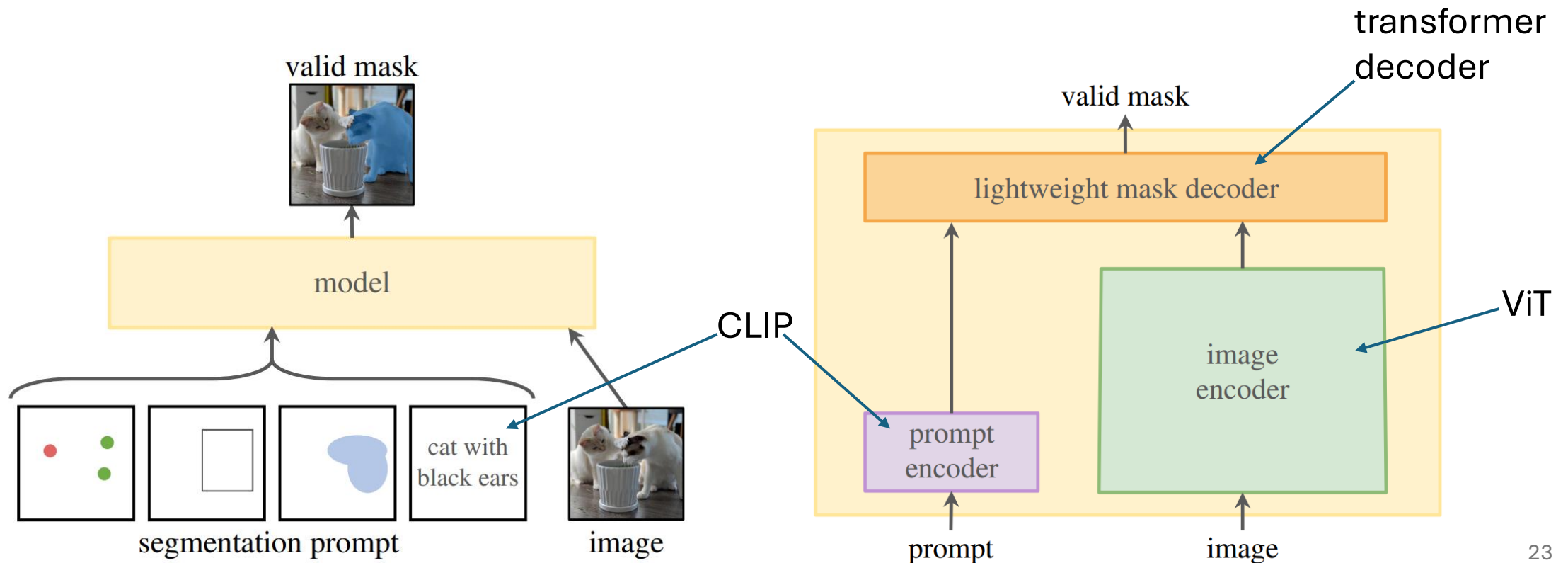
Detectron2: PyTorch Implementations



Promptable Segmentation with Transformers

Segment Anything Model ([SAM](#))

[SAM2](#) includes also video segmentation



Some Off-the-Shelf ML Methods

multivariate tabular data prediction: HistGradientBoosting from scikit-learn ([regression](#), [classification](#))

image classification: feature extraction from [DINOv2](#) or finetune a ResNet/ViT from [torchvision](#) (zero-shot: [CLIP](#))

object detection: Faster R-CNN ResNet-50 FPN from torchvision (zero-shot: [YOLOv8](#))

semantic segmentation: FCN ResNet-50 from torchvision (zero-shot: YOLOv8-seg)

instance segmentation: Mask R-CNN ResNet-50 FPN from torchvision (zero-shot: YOLOv8-seg or [SAM](#))

text classification: finetune [DistilBERT](#) from transformers (zero-shot: prompt an [Ollama](#) LLM)